

VirtualWorks Lab Internship

Task 4 — Case Classification & Severity Analysis

Simulated Pharmacovigilance Case Dataset

About this submission

This task applies the case-classification concept from the assigned tutorial to a simulated pharmacovigilance dataset. Each case is assigned structured demographic and clinical attributes so that cases can be grouped and compared. The exercise then uses those classifications for basic severity, seriousness, expectedness, age-group, and ADR-category analysis.

Disclaimer

All cases, patient characteristics, medicines, adverse events, dates and outcomes in this report are fictional/simulated and were created only for internship training. They do not represent real patients, real safety reports, or a validated pharmacovigilance database.

Classification framework

Classification attribute	Values used
Age Group	Pediatric / Adult / Elderly
Sex	Female / Male
ADR Category	Dermatological / Gastrointestinal / Cardiovascular / Hepatic / Musculoskeletal / Neurological / Respiratory
Severity	Mild / Moderate / Severe
Seriousness	Serious / Non-serious
Expectedness	Expected / Unexpected
Outcome	Recovered / Recovering

Simulated Case Classification Dataset

Case ID	Age	Sex	Drug	ADR / Event	Category	Severity	Seriousness	Expectedness	Outcome
CC-001	24	Female	Amoxicillin	Generalized urticaria and itching	Dermatological / Hypersensitivity	Mild	Non-serious	Expected	Recovered
CC-002	67	Male	Amlodipine	Bilateral ankle edema	Cardiovascular / Fluid-related	Moderate	Non-serious	Expected	Recovered
CC-003	45	Female	Metformin	Diarrhea and abdominal discomfort	Gastrointestinal	Mild	Non-serious	Expected	Recovered
CC-004	58	Male	Methotrexate	Elevated liver enzymes with fatigue	Hepatic / Laboratory abnormality	Moderate	Serious	Expected	Recovering
CC-005	31	Female	Carbamazepine	Severe cutaneous reaction with mucosal involvement	Dermatological / Severe cutaneous	Severe	Serious	Expected	Recovered
CC-006	72	Male	Warfarin	Gastrointestinal bleeding requiring hospitalization	Gastrointestinal / Hemorrhagic	Severe	Serious	Expected	Recovered
CC-007	39	Female	Lisinopril	Persistent dry cough	Respiratory	Mild	Non-serious	Expected	Recovered
CC-008	63	Male	Atorvastatin	Muscle pain with elevated CK	Musculoskeletal	Moderate	Non-serious	Expected	Recovering
CC-009	52	Female	Levofloxacin	Tendon rupture requiring surgery	Musculoskeletal / Tendon injury	Severe	Serious	Expected	Recovering
CC-010	28	Male	Sertraline	Agitation and insomnia	Neurological / Psychiatric	Mild	Non-serious	Expected	Recovered
CC-011	61	Female	Allopurinol	Facial swelling and breathing difficulty	Dermatological / Hypersensitivity	Severe	Serious	Unexpected	Recovered
CC-012	49	Male	Omeprazole	Acute confusion and dizziness	Neurological	Moderate	Non-serious	Unexpected	Recovered

The Excel workbook contains the full classification table, including the derived Age Group field, a data dictionary, classification guide, and summary charts.

Classification & Analysis Findings

- **Severity:** 4 mild, 4 moderate, and 4 severe cases were represented in the simulated dataset.
- **Seriousness:** 5 cases were classified as serious and 7 as non-serious.
- **Expectedness:** 10 events were classified as expected and 2 as unexpected within the exercise.
- **Age groups:** 10 cases were classified as adult and 2 as elderly; no pediatric cases were included.
- **ADR categories:** The simulated cases span dermatological/hypersensitivity, gastrointestinal, cardiovascular/fluid-related, hepatic/laboratory, musculoskeletal, neurological/psychiatric, and respiratory categories.

Why case classification matters

Structured classification converts individual case records into comparable groups. For example, severity can be reviewed across age groups, seriousness can be compared with ADR categories, and unexpected events can be isolated for closer review. The exercise therefore demonstrates how a classification table can support organized pharmacovigilance data review rather than treating each case as an isolated narrative.

Limitations

This is a small simulated dataset created for learning. The classifications are intentionally simplified and should not be interpreted as a real-world safety signal, incidence estimate, clinical conclusion, or regulatory assessment. No statistical inference or signal-detection conclusion is intended.

Learning objective

This task helped me understand how pharmacovigilance cases can be organized using structured classification attributes and then summarized to identify patterns across patient and event characteristics. It also reinforced the importance of keeping simulated training data clearly separated from real patient safety information.

Deliverables

- Task_4_Case_Classification_Severity_Analysis.pdf — case classification report
- Task_4_Case_Classification_Severity_Analysis.xlsx — full simulated dataset, classification guide, data dictionary, and summary analysis